

Lesson 6: Interpretations and Visualizations of Odds Ratios

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Learning Objectives

1. Interpret odds ratios from fitted simple logistic regression model for a continuous explanatory variable.
2. Interpret odds ratios from fitted simple logistic regression model for a binary explanatory variable.
3. Interpret odds ratios from fitted simple logistic regression model for a multi-level categorical explanatory variable.
4. Report the odds ratio using a table and/or a forest plot.

Recall our example: Late stage breast cancer diagnosis

- Recall that we fit ~~the~~ a simple logistic regression for late stage breast cancer diagnosis using the predictor, age:

```
1 bc_reg = bc %>% glm(formula = Late_stage_diag ~ Age_c, family = binomial)
2 tidy(bc_reg, conf.int=T) %>% gt() %>% tab_options(table.font.size = 38) %>%
3   fmt_number(decimals = 3)
```

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	-0.989	0.023	-42.637	0.000	-1.035	-0.944
Age_c	0.057	0.003	17.780	0.000	0.051	0.063

Population logistic regression model

$$\text{logit}(\pi(\text{Age})) = \beta_0 + \beta_1 \cdot \text{Age}$$

$$\log\left(\frac{\pi(\text{Age})}{1 - \pi(\text{Age})}\right)$$

$$\pi(\text{Age}) = P(Y=1 | \text{Age})$$

Fitted logistic regression model

$$\text{logit}(\hat{\pi}(\text{Age})) = -0.989 + 0.057 \cdot \text{Age}$$

$\hat{\beta}_0$

$\hat{\beta}_1$

Coefficients on the log-odds scale

Population logistic regression model

$$\text{logit}(\pi(X)) = \beta_0 + \beta_1 \cdot X$$

"Successful" outcome

- β_0 : log-odds of $Y = 1$ when X is 0
- β_1 : increase in log-odds of $Y = 1$ for every 1 unit increase in X

Fitted logistic regression model

$$\text{logit}(\hat{\pi}(X)) = \hat{\beta}_0 + \hat{\beta}_1 \cdot X$$

- $\hat{\beta}_0$: estimated log-odds of $Y = 1$ when X is 0
- $\hat{\beta}_1$: estimated increase in log-odds of $Y = 1$ for every 1 unit increase in X
- Can use expected instead of estimated

Recall our example: Late stage breast cancer diagnosis

- Fitted logistic regression model:

$$\text{logit}(\hat{\pi}(\text{Age})) = -0.989 + 0.057 \cdot \text{Age}$$

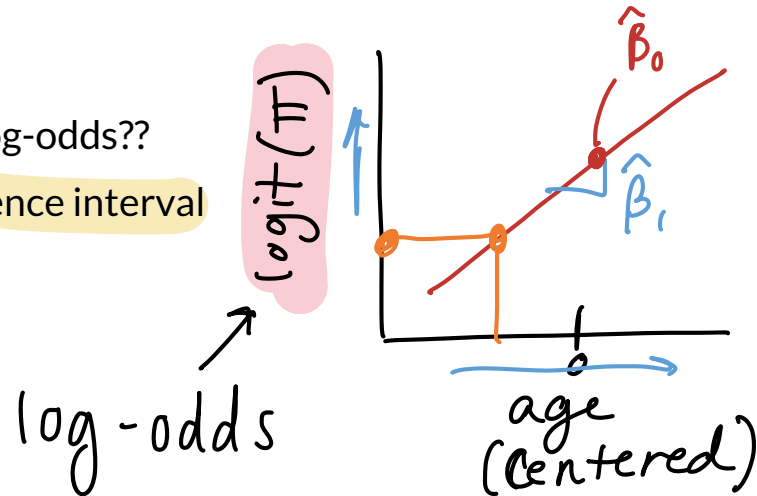
$\hat{\beta}_0$ $\hat{\beta}_1$

- $\hat{\beta}_0$: The estimated log-odds is -0.989 for someone who is 61.71 years (95% CI: -1.035, -0.944)
- $\hat{\beta}_1$: The estimated increase in log-odds is 0.057 for every 1 year increase in age (95% CI: 0.051, 0.063).

mean age - bk age is centered this is the "0"

$\text{Age}^c = 0 = 61.71$
- mean age

- What does a log-odds even mean?? Or an increase in log-odds??
- We will need to calculate the odds ratio and its confidence interval
 - Then we will visualize the odds ratio



Poll Everywhere Question 1

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...

75%

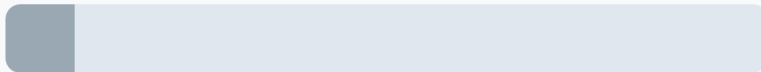


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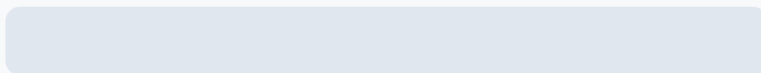
How do we transform our coefficient estimates to get the estimated odds ratios?

No transformation needed



5%

Take the difference between the estimates



0%

Take the log of the estimates



45%

Take the exponential of the estimates ✓



50%

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→ log odds
↳ log log odds
~~exp(log())~~

We typically interpret our results using odds ratios

For our *fitted* simple logistic regression model with a continuous predictor

$$\longrightarrow \text{logit}(\hat{\pi}(X)) = \hat{\beta}_0 + \hat{\beta}_1 \cdot X$$

- How do we go from interpretations of $\hat{\beta}_0$ and $\hat{\beta}_1$ using log odds to odds ratios?

- We will need to take the exponential of our model:

- $\exp(\hat{\beta}_0)$: expected odds that $Y = 1$ when X is 0.
- $\exp(\hat{\beta}_1)$: expected odds ratio that $Y = 1$ for every 1 unit increase in X

- Important distinction:

- We take the inverse logit to find our predicted probability (next lesson)
- We take the exponential to interpret the odds/odds ratios

Intro/Recap of Interpreting Fitted Model

- Interpret coefficients from fitted logistic regression model
 - **Goodness-of-fit of model should be assessed before summarizing findings** (have not covered yet)
 - In this lecture: assume model fits data well
- The interpretation of the coefficients involves two issues:
 - The functional relationship between the dependent variable and the independent variable (*link function*)
 - Unit of change for the independent variable
- We will learn the interpretation for
 - Binary independent variable
 - Categorical independent variable with multiple groups
 - We looked at this for our race and ethnicity variable
 - Continuous independent variable

Learning Objectives

1. Interpret odds ratios from fitted simple logistic regression model for a continuous explanatory variable.
2. Interpret odds ratios from fitted simple logistic regression model for a binary explanatory variable.
3. Interpret odds ratios from fitted simple logistic regression model for a multi-level categorical explanatory variable.
4. Report the odds ratio using a table and/or a forest plot.

Coefficient interpretation: Continuous Independent Variable

- For simplicity, we assume the linear relationship between logit and continuous variable X
- Again using simple logistic regression model to illustrate the interpretation of $\hat{\beta}$ for a continuous variable X

$$\text{logit}(\hat{\pi}(X)) = \hat{\beta}_0 + \hat{\beta}_1 \cdot X$$

- The estimated slope coefficient, $\hat{\beta}_1$, is the **expected change in the log odds for 1 unit increase in X**
 - Additional attention should be paid to picking a meaningful units of change in X

How do we get the odds ratio for X's coefficient?

$$\text{logit}(k) = \log\left(\frac{k}{1-k}\right)$$

For $\exp(\hat{\beta}_1)$

• Thus,

- We compare $X = x$ and $X = x + 1$

- So we have

$$\text{logit}(\hat{\pi}(X = x)) = \hat{\beta}_0 + \hat{\beta}_1 \cdot x$$

and

$$\text{logit}(\hat{\pi}(X = x + 1)) = \hat{\beta}_0 + \hat{\beta}_1 \cdot (x + 1)$$

- And...

$$\text{logit}(\hat{\pi}(X = x + 1)) - \text{logit}(\hat{\pi}(X = x))$$

$$= (\hat{\beta}_0 + \hat{\beta}_1 \cdot (x + 1)) - (\hat{\beta}_0 + \hat{\beta}_1 \cdot x)$$

$$= \hat{\beta}_0 + \hat{\beta}_1 \cdot x + \hat{\beta}_1 - \hat{\beta}_0 - \hat{\beta}_1 \cdot x$$

$$= \hat{\beta}_1$$

$$\hat{\beta}_1 = \text{logit}(\hat{\pi}(X = x + 1)) - \text{logit}(\hat{\pi}(X = x))$$

$$\hat{\beta}_1 = \log\left(\frac{\hat{\pi}(X = x + 1)}{1 - \hat{\pi}(X = x + 1)}\right) - \log\left(\frac{\hat{\pi}(X = x)}{1 - \hat{\pi}(X = x)}\right)$$

$$\hat{\beta}_1 = \log\left(\frac{\frac{\hat{\pi}(X = x + 1)}{1 - \hat{\pi}(X = x + 1)}}{\frac{\hat{\pi}(X = x)}{1 - \hat{\pi}(X = x)}}\right)$$

$$\log(a) - \log(b) = \log\left(\frac{a}{b}\right)$$

$$\exp[\hat{\beta}_1] = \exp\left[\log\left(\frac{\widehat{\text{odds}}_{X=x+1}}{\widehat{\text{odds}}_{X=x}}\right)\right]$$

$$\exp[\hat{\beta}_1] = \frac{\widehat{\text{odds}}_{X=x+1}}{\widehat{\text{odds}}_{X=x}} \rightarrow \widehat{OR} \text{ for } Y=1$$

comparing $x+1$ to x (1-unit inc in X)

Example: Interpretation of Age Coefficient/OR

- $\hat{\beta}_1$ is 0.057, suggesting that one year increase in age is associated with 0.057 increase in log odds of receiving a late stage breast cancer diagnosis

- $\exp(\hat{\beta}_1)$ is 1.06, suggesting that one year increase in age is associated with 1.06 times the odds of receiving a late stage breast cancer diagnosis

- Can also say: For every one year increase in age, the odds of late stage breast cancer diagnosis is 1.06 times.

WRONG "odds increase by 1.06" $\rightarrow$ if x : odds = 0.5
 $x+1$: odds = $0.5 + 1.06 = 1.56$

- For continuous covariates in logistic regression model, it is helpful to subtract 1 from the odds ratio and multiply by 100 to obtain the percentage change in odds for 1-unit increase.

- The estimated OR for age is 1.06, suggesting that a 1-year increase in age is associated with a 6% increase in the predicted odds of late stage diagnosis

RIGHT: "odds 1.06 times increase" if x : odds = 0.5
if $x+1$: odds = $1.06 \cdot 0.5$
 $= 0.53$

Example: Age and Late Stage Diagnosis (1/5)

Odds ratio from logistic regression

Compute the estimate and 95% confidence interval for odds ratio ^{of} late stage breast cancer diagnosis for every 1 year increase in age.

Needed steps:

1. Fit the regression model
2. Transform the coefficients into odds ratios
3. Interpret the odds ratio

Example: Age and Late Stage Diagnosis (2/5)

Odds ratio from logistic regression

Compute the estimate and 95% confidence interval for odds ratio for late stage breast cancer diagnosis for every 1 year increase in age.

1. Fit the regression model

```
1 bc_reg = bc %>%  
2   glm(formula = Late_stage_diag ~ Age_c, family = binomial)
```

Example: Age and Late Stage Diagnosis (3/5)

Odds ratio from logistic regression

Compute the estimate and 95% confidence interval for odds ratio for late stage breast cancer diagnosis for every 1 year increase in age.

2. Transform the coefficients into odds ratios • Option 1: `tidy()`

```
1 tidy_bc_reg = tidy(bc_reg, conf.int=T, exponentiate = T)
2 tidy_bc_reg %>% gt() %>% tab_options(table.font.size = 35) %>%
3   fmt_number(decimals = 3)
```

takes you from β_1 to $\exp(\hat{\beta}_1)$

$\exp(\hat{\beta}_0)$
odds when $x=0$

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	0.372	0.023	-42.637	0.000	0.355	0.389
Age_c	1.059	0.003	17.780	0.000	1.052	1.065

```
1 tidy_bc_reg$conf.low # I prefer tidy() bc now I can grab each component
```

```
[1] 0.3551931 1.0520321
```

Example: Age and Late Stage Diagnosis (4/5)

Odds ratio from logistic regression

Compute the estimate and 95% confidence interval for odds ratio for late stage breast cancer diagnosis for every 1 year increase in age.

2. Transform the coefficients into odds ratios • Option 2: `logistic.display()`

```
1 logistic.display(bc_reg) # Cannot grab each component in this
```

Logistic regression predicting Late_stage_diag : 1 vs 0

	OR(95%CI)	P(Wald's test)	P(LR-test)
Age_c (cont. var.)	1.06 (1.05, 1.07)	< 0.001	< 0.001

Log-likelihood = -5754.8442
No. of observations = 10000
AIC value = 11513.6884

Handwritten notes: An arrow points from the OR(95%CI) column to the value 1.00 in the P(Wald's test) column. The value 1.00 is circled.

Poll Everywhere Question 2

13:45 Mon Apr 14

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Please fill in the blanks for the following interpretation of our $\hat{OR}$: A _____ year increase in age is associated with _____ (95% CI: (1.05, 1.07)) times the odds of a late stage breast cancer diagnosis.

10 year, 1.06		0%
1 year, 1.06 ✓		95%
10 year, 0.06		0%
1 year, 0.06		5%

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→ % inc
could 6%

Example: Age and Late Stage Diagnosis (5/5)

Odds ratio from logistic regression

Compute the estimate and 95% confidence interval for odds ratio for late stage breast cancer diagnosis for every 1 year increase in age.

3. Interpret the odds ratio

For every one year increase in age, there is an expected 5.86% increase in the odds of late stage breast cancer diagnosis (95% CI: 5.2%, 6.53%).

Transformations of continuous variable to make more interpretable

- Sometimes a change in “1” unit may not be considered clinically interesting
- For example, a 1 year increase in age or a 1 mm Hg increase in systolic blood pressure may be too small for a meaningful change in log odds
 - Instead, we may be interested to find out the log odds change for a increase of 10 years in age or 10 mm Hg in systolic blood pressure
- On the other hand, if the range of X is small (say 0-1), then a change in 1 unit of X is too large to be meaningful
- We should be able to compute and interpret coefficients for a continuous independent covariate x for an arbitrary change of “c” units in x

Transformations of continuous variable to make more interpretable

- The estimated log odds ratio for a change of c units in x can be obtained from

$$\hat{g}(x + c) - \hat{g}(x) = c\hat{\beta}_1 \quad 10^{\hat{\beta}_1}$$

$0.01 \rightarrow 0.01\hat{\beta}_1$

$$\widehat{OR}(c) = \exp(c\hat{\beta}_1)$$

- The 95% CI for $\widehat{OR}(c)$ is:

$$\exp\left(c\hat{\beta}_1 \pm 1.96 \cdot c \cdot SE_{\hat{\beta}_1}\right)$$

$$\left(\exp\left(\frac{c\hat{\beta}_1 - 1.96cSE}{1}\right), \exp\left(\frac{c\hat{\beta}_1 + 1.96cSE}{1}\right) \right)$$

- The c is chosen to be a clinically meaningful unit change in x
- The value of c should be clearly specified in all tables and calculations
 - Because the estimated OR and the corresponding CI depends on the choice of c value

Example: 10 year increase in age and Late Stage Diagnosis

- What if we are interested in learning the OR corresponding to 10-year increase in age?

```
1 bc_reg = glm(Late_stage_diag ~ Age_c, data = bc, family = binomial)
2 tidy(bc_reg, conf.int=T) %>% gt() %>% tab_options(table.font.size = 35) %>%
3   fmt_number(decimals = 3)
```

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	-0.989	0.023	-42.637	0.000	-1.035	-0.944
Age_c	0.057	0.003	17.780	0.000	0.051	0.063

Example: 10 year increase in age and Late Stage Diagnosis

- What if we are interested in learning the OR corresponding to 10-year increase in age?

$$\widehat{OR}(10) = \exp(10 \cdot \hat{\beta}_1) = \exp(0.56965) = 1.767$$

10.0.057

- The 95% CI for $\widehat{OR}(10)$ is:

$$\begin{aligned}\widehat{OR}(10) &= \exp\left(10 \cdot \hat{\beta}_1 \pm 1.96 \cdot 10 \cdot SE_{\hat{\beta}_1}\right) \\ &= \exp(10 \cdot 0.056965 \pm 1.96 \cdot 10 \cdot 0.003204) \\ &= (1.66, 1.88)\end{aligned}$$

expected
10 year inc in age, odds of LSBC diagnosis
is 1.767 times.

Example: 10 year increase in age and Late Stage Diagnosis

- What if we are interested in learning the OR corresponding to 10-year increase in age?

```
1 bc2 = bc %>% mutate(Age_c_10 = Age_c/10) | unit inc = 10 yr inc
2 bc_reg_10 = glm(Late_stage_diag ~ Age_c_10, data = bc2, family = binomial)
3 tidy(bc_reg_10, conf.int=T, exponentiate = T) %>% gt() %>% tab_options(table.font.size = 35)
4   fmt_number(decimals = 3)
```

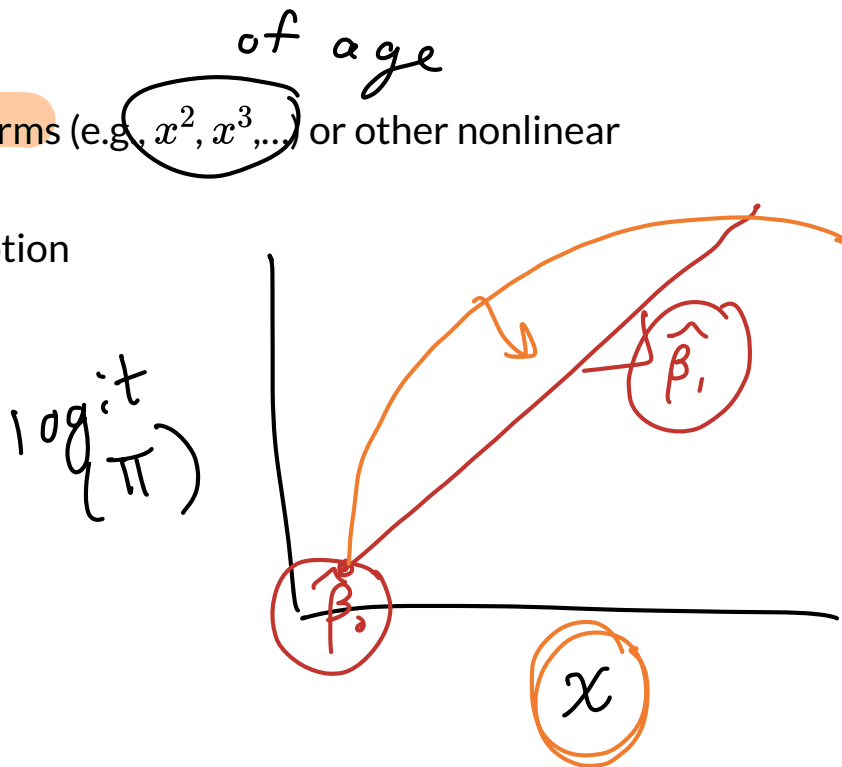
odds &
ORs

	term	estimate	std.error	statistic	p.value	conf.low	conf.high
$\hat{odds}$ →	(Intercept)	0.372	0.023	-42.637	0.000	0.355	0.389
$\hat{OR}$ →	Age_c_10	1.768	0.032	17.780	0.000	1.661	1.883

Last Note About Continuous Independent Variable

- Notice that the logistic regression model suggests that logit is linear in the covariate
- The model implies the additional ~~risk~~^{odds} of late stage breast cancer diagnosis for a 40 year-old compared to a 30 year-old is the same as the additional ~~risk~~^{odds} of late stage breast cancer diagnosis for a 60 year-old compared to a 50-year-old
- This assumption may not be realistic
- To address this, we may consider using higher order terms (e.g., $x^2, x^3, \dots$) or other nonlinear transformation (e.g., $\log(x)$) of age
- Categorize the continuous variable may be another option

* add ref to
Linear Models



How do we get the odds for the intercept?

For $\exp(\hat{\beta}_0)$

- When $X = 0$, we have

$$\text{logit}(\hat{\pi}(X=0)) = \hat{\beta}_0 + \hat{\beta}_1(0)$$

- Thus,

$$\hat{\beta}_0 = \text{logit}(\hat{\pi}(X)) \quad \xrightarrow{X=0} \quad = 0$$

$$\exp[\hat{\beta}_0] = \exp[\text{logit}(\hat{\pi}(X))]$$

$$\exp[\hat{\beta}_0] = \exp\left[\log\left(\frac{\hat{\pi}(X)}{1 - \hat{\pi}(X)}\right)\right]$$

$$\exp[\hat{\beta}_0] = \frac{\hat{\pi}(X \neq 0)}{1 - \hat{\pi}(X \neq 0)} \quad \checkmark \text{ odds of } Y=1 \text{ when } X=0 \text{ expected}$$

Learning Objectives

1. Interpret odds ratios from fitted simple logistic regression model for a continuous explanatory variable.
2. Interpret odds ratios from fitted simple logistic regression model for a binary explanatory variable.
3. Interpret odds ratios from fitted simple logistic regression model for a multi-level categorical explanatory variable.
4. Report the odds ratio using a table and/or a forest plot.

Coefficient Interpretation: Binary Independent Variable

- Independent variable X is a binary variable (X can take values: 0 or 1)

population

- We are fitting the simple logistic regression model:

$$\text{logit}(\pi(X)) = \beta_0 + \beta_1 \cdot I(X = 1)$$

- The logit difference is β_1 for binary independent variable
 - β_1 represents the change/difference in the logit for $X = 1$ vs. $X = 0$
- It will be much easier to understand if we can interpret the coefficient using odds ratio (OR)

Binary: How do we interpret the coefficient? (I)

- For individuals with $X = 0$:

$$\text{logit}(\pi(X = 0)) = \beta_0 + \beta_1 \times (0) = \beta_0$$

$$\hookrightarrow I(X = 1) = 0$$

- For individuals with $X = 1$:

$$\text{logit}(\pi(X = 1)) = \beta_0 + \beta_1 \times (1) = \beta_0 + \beta_1$$

$$\hookrightarrow I(X = 1) = 1$$

"yes", "exposed", "trt", etc.

- To solve for β_1 , we take the difference of the logits:

$$\text{logit}(\pi(X = 1)) - \text{logit}(\pi(X = 0)) = (\beta_0 + \beta_1) - (\beta_0) = \beta_1$$

Binary: How do we interpret the coefficient? (II)

$$\text{logit}(\pi(X=1)) - \text{logit}(\pi(X=0)) = (\beta_0 + \beta_1) - (\beta_0) = \beta_1$$

$$\beta_1 = \text{logit}(\pi(X=1)) - \text{logit}(\pi(X=0))$$

$$\beta_1 = \log\left(\frac{\pi(X=1)}{1 - \pi(X=1)}\right) - \log\left(\frac{\pi(X=0)}{1 - \pi(X=0)}\right)$$

$$\beta_1 = \log\left(\frac{\frac{\pi(X=1)}{1 - \pi(X=1)}}{\frac{\pi(X=0)}{1 - \pi(X=0)}}\right)$$

$\beta_1 = \text{log-odds ratio}$
of $Y=1$ comparing
- $X=1$ to $X=0$

$$\exp(\beta_1) = \frac{\frac{\pi(X=1)}{1 - \pi(X=1)}}{\frac{\pi(X=0)}{1 - \pi(X=0)}}$$

$\exp(\beta_1) = \text{odds ratio}$
of $Y=1$ comparing
- $X=1$ to $X=0$

Review of Odds Ratio

- Odds for a subject with $X = 1$:

$$\text{odds}_1 = \frac{\pi(X = 1)}{1 - \pi(X = 1)}$$

- Odds for a subject with $X = 0$:

$$\text{odds}_0 = \frac{\pi(X = 0)}{1 - \pi(X = 0)}$$

- Odds Ratio for $X = 1$ vs. $X = 0$:

$$OR = \frac{\frac{\pi(X = 1)}{1 - \pi(X = 1)}}{\frac{\pi(X = 0)}{1 - \pi(X = 0)}}$$

How does this relate to a 2x2 table?

- 2x2 table with the respective logistic functions in each cell

Outcome Variable (Y)	Independent Variable (X)	
	X = 1	X = 0
Y = 1	$\pi(1) = \frac{\exp(\beta_0 + \beta_1)}{1 + \exp(\beta_0 + \beta_1)}$	$\pi(0) = \frac{\exp(\beta_0)}{1 + \exp(\beta_0)}$
Y = 0	$1 - \pi(1) = \frac{1}{1 + \exp(\beta_0 + \beta_1)}$	$1 - \pi(0) = \frac{1}{1 + \exp(\beta_0)}$
Total	1.0	1.0

Recall:

$$\pi(1) = \pi(X = 1) \\ = P(Y = 1 | X = 1)$$

$$\pi(0) = \pi(X = 0) \\ = P(Y = 1 | X = 0)$$

$$P(Y=1 | X=1) \\ + P(Y=0 | X=1) \\ = 1$$

Poll Everywhere Question 3

oops

Example: Binary age and Late Stage Diagnosis (1/4)

Odds ratio from logistic regression

What is the odds ratio of late stage breast cancer diagnosis for older individuals (>65 years old) compared to younger individuals (≤ 65 years old)?

- Two options to calculate this value:

- **Option 1:** Calculate $\widehat{OR}$ from 2x2 contingency table ✓
 - Try at home: Refer to [Lesson 3](#) for this process
- **Option 2:** Calculate $\widehat{OR}$ from logistic regression ✓

Needed steps for **Option 2**:

1. Fit the regression model
2. Transform the coefficients into odds ratios
3. Interpret the odds ratio

Example: Binary age and Late Stage Diagnosis (2/4)

Odds ratio from logistic regression

What is the odds ratio of late stage breast cancer diagnosis for older individuals (>65 years old) compared to younger individuals (≤ 65 years old)?

1. Fit the regression model

```
1 bc3 = bc %>% mutate(Age_binary = ifelse(Age > 65, 1, 0))
2 age_bin_glm = bc3 %>% glm(formula = Late_stage_diag ~ Age_binary, family = binomial)
```

Example: Binary age and Late Stage Diagnosis (3/4)

Odds ratio from logistic regression

What is the odds ratio of late stage breast cancer diagnosis for older individuals (>65 years old) compared to younger individuals (≤ 65 years old)?

2. Transform the coefficients into odds ratios

```
1 age_bin_tidy = tidy(age_bin_glm, conf.int=T, exponentiate = T)
2 age_bin_tidy %>% gt() %>%
3   tab_options(table.font.size = 35) %>%
4   fmt_number(decimals = 3)
```

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	0.297	0.031	-39.608	0.000	0.280	0.315
Age_binary	1.875	0.045	13.928	0.000	1.716	2.048

Poll Everywhere Question 4

14:24 Mon Apr 14

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Please fill in the blanks for the following interpretation of our $\hat{OR}$: The estimated odds of late stage breast cancer among individuals over 65 years old is 1.87 (95% CI: (_____, 2.05)) times _____ than individuals 65 years or younger.

Higher → that of

0.29, higher	0%
0.29, lower	0%
1.72, higher ✓	100%
1.72, lower	0%

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Example: Binary age and Late Stage Diagnosis (4/4)

Odds ratio from logistic regression

What is the odds ratio of late stage breast cancer diagnosis for older individuals (>65 years old) compared to younger individuals (≤ 65 years old)?

3. Interpret the odds ratio

The estimated odds of late stage breast cancer among individuals over 65 years old is 1.87 (95% CI: (1.72, 2.05)) times that of individuals 65 years or younger.

Learning Objectives

1. Interpret odds ratios from fitted simple logistic regression model for a continuous explanatory variable.
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4. Report the odds ratio using a table and/or a forest plot.

Coefficient Interpretation: Multi-group Categorical Variable

- Independent variable X is a multi-level categorical variable

- Let's say X takes values: a, b, c, or d

- We are fitting the simple logistic regression model:

$$\text{logit}(\pi(X)) = \beta_0 + \beta_1 \cdot I(X = b) + \beta_2 \cdot I(X = c) + \beta_3 \cdot I(X = d)$$

- Where a is our reference group

systematic comp is same
as Linear Models

- β_1 represents the change/difference in the log-odds for $X = b$ vs. $X = a$

Coefficient Interpretation: Multi-group Categorical Variable

We are fitting the simple logistic regression model with reference group a :

$$\text{logit}(\pi(X)) = \beta_0 + \beta_1 \cdot I(X = b) + \beta_2 \cdot I(X = c) + \beta_3 \cdot I(X = d)$$

- β_0 : the log-odds of event $Y = 1$ for group a

$\downarrow$
 $= 0$

$\downarrow$
 $= 1$

$\downarrow$
 $= 0$

$\swarrow$ if in grp c

- β_1 : the difference in log-odds of event $Y = 1$ comparing group b to group a

- β_2 : the difference in log-odds of event $Y = 1$ comparing group c to group a

- β_3 : the difference in log-odds of event $Y = 1$ comparing group d to group a

Multi-level categorical: How do we interpret the coefficient? (II)

$$\text{logit}(\pi(X=c)) - \text{logit}(\pi(X=a)) = (\beta_0 + \beta_1 \cdot 0 + \beta_2 \cdot 1 + \beta_3 \cdot 0) - (\beta_0 + \beta_1 \cdot 0 + \beta_2 \cdot 0 + \beta_3 \cdot 0) = \beta_2$$

$$\beta_2 = \text{logit}(\pi(X=c)) - \text{logit}(\pi(X=a))$$

$$\beta_2 = \log\left(\frac{\pi(X=c)}{1 - \pi(X=c)}\right) - \log\left(\frac{\pi(X=a)}{1 - \pi(X=a)}\right)$$

$$\beta_2 = \log\left(\frac{\frac{\pi(X=c)}{1 - \pi(X=c)}}{\frac{\pi(X=a)}{1 - \pi(X=a)}}\right)$$

$$\exp(\beta_2) = \frac{\frac{\pi(X=c)}{1 - \pi(X=c)}}{\frac{\pi(X=a)}{1 - \pi(X=a)}}$$

odds of $Y=1$ for grp c
compared to odds of
 $Y=1$ for grp a

Coefficient Interpretation: Multi-group Categorical Variable

We are fitting the simple logistic regression model with reference group a :

$$\text{logit}(\pi(X)) = \beta_0 + \beta_1 \cdot I(X = b) + \beta_2 \cdot I(X = c) + \beta_3 \cdot I(X = d)$$

- $\exp(\beta_0)$: the odds of event $Y = 1$ for group a
 - $\exp(\beta_1)$: the odds of event $Y = 1$ for group b is $\exp(\beta_1)$ times the odds of event $Y = 1$ for group a
 - $\exp(\beta_2)$: the odds of event $Y = 1$ for group c is $\exp(\beta_2)$ times the odds of event $Y = 1$ for group a
 - $\exp(\beta_3)$: the odds of event $Y = 1$ for group d is $\exp(\beta_3)$ times the odds of event $Y = 1$ for group a
-
- Remember, as soon as we fit the regression model, we talk about the “expected odds” or “estimated odds”

How do we pick the reference group?

- The choice can be more apparent for multi-group categorical independent variables within studies
- For example, if we want to evaluate the association between clinical response and four treatments.
 - The treatment variable has 4 categories: “active treatment A”, “active treatment B”, “active treatment C” and “Placebo treatment”
 - The investigator is interested in comparing each of the three active treatment with the placebo treatment
 - Then the placebo treatment should be picked as the reference group

Example: Late stage diagnosis and race and ethnicity

- We chose Non-Hispanic White individuals as reference group

- Underlying health disparities linked to racism in healthcare and in clinical studies

- There is evidence that white individuals receive a certain standard of care that is not paralleled for POC (Mateo and Williams (2021))

- Something else to consider: if your reference group does not have a lot of individuals, then you may not see any statistical significance!

Race/Ethnicity	Breast Cancer Diagnosis		Total
	Early Stage	Late Stage	
Non-Hispanic White	5,321	1,980	7,301
Non-Hispanic Black	683	357	1,040
Non-Hispanic Asian/Pacific Islander	556	234	790
Hispanic-Latinx	575	271	846
Non-Hispanic American Indian/Alaska Native	17	6	23
Total	7,152	2,848	10,000

Example: Late stage diagnosis and race and ethnicity

Odds ratio from logistic regression

What is the odds ratio of late stage breast cancer diagnosis for Non-Hispanic Asian/Pacific Islander individuals compared to Non-Hispanic White individuals?

Needed steps:

1. Fit the regression model
2. Transform the coefficients into odds ratios
3. Interpret the odds ratio

Example: Late stage diagnosis and race and ethnicity

Odds ratio from logistic regression

What is the odds ratio of late stage breast cancer diagnosis for Non-Hispanic Asian/Pacific Islander individuals compared to Non-Hispanic White individuals?

1. Fit the regression model

```
1 RE_glm = bc %>%  
2   glm(formula = Late_stage_diag ~ Race_Ethnicity, family = binomial)
```

$$\text{logit}(\pi(X)) = \beta_0 + \beta_1 I(\text{RE} = \text{"NH Black"}) + \beta_2 I(\text{RE} = \text{"^{NH}Asian/Pacific Islander"}) + \beta_3 I(\text{RE} = \text{"Hispanic/Latinx"}) + \beta_4 I(\text{RE} = \text{"NH American Indian/Alaskan Native"})$$

Example: Late stage diagnosis and race and ethnicity

Odds ratio from logistic regression

What is the odds ratio of late stage breast cancer diagnosis for Non-Hispanic Asian/Pacific Islander individuals compared to Non-Hispanic White individuals?

2. Transform the coefficients into odds ratios

```
1 RE_tidy = tidy(RE_glm, conf.int=T, exponentiate = T)
2 RE_tidy %>% gt() %>%
3   tab_options(table.font.size = 35) %>%
4   fmt_number(decimals = 3)
```

odds/
OR

	term	estimate	std.error	statistic	p.value	conf.low	conf.high
$\exp \hat{\beta}_0$	(Intercept)	0.372	0.026	-37.553	0.000	0.353	0.392
$\exp \hat{\beta}_1$	Race_EthnicityHispanic-Latino	0.968	0.082	-0.398	0.691	0.822	1.135
$\exp \hat{\beta}_2$	Race_EthnicityNH American Indian/Alaskan Native	0.948	0.476	-0.111	0.911	0.342	2.287
$\exp \hat{\beta}_3$	Race_EthnicityNH Asian/Pacific Islander	1.131	0.082	1.497	0.134	0.961	1.327
$\exp \hat{\beta}_4$	Race_EthnicityNH Black	1.405	0.070	4.826	0.000	1.223	1.611

Example: Late stage diagnosis and race and ethnicity

Odds ratio from logistic regression

What is the odds ratio of late stage breast cancer diagnosis for Non-Hispanic Asian/Pacific Islander individuals compared to Non-Hispanic White individuals?

3. Interpret the odds ratio

The estimated odds of late stage breast cancer among Non-Hispanic Asian/Pacific Islander individuals is 1.13 (95% CI: (0.96, 1.33)) times that of Non-Hispanic White individuals.

What if you want to compare other groups?

- What if we want to estimate OR comparing Non-Hispanic Asian Pacific Islander to Non-Hispanic Black individuals?
- **Option 1:** Change reference group and refit the model (maybe the easiest option)
- **Option 2:** Estimate OR using fitted coefficients ($\hat{\beta}$'s) in the current model:

$$\begin{aligned}\log(OR(\text{NH API}, \text{NH B})) &= \text{logit}(\pi(X = \text{NH API})) - \text{logit}(\pi(X = \text{NH B})) \\ &= [\beta_0 + \beta_3(1)] - [\beta_0 + \beta_4(1)] \\ \log(\widehat{OR}(\text{NH API}, \text{NH B})) &= \hat{\beta}_3 - \hat{\beta}_4 \\ \widehat{OR}(\text{NH API}, \text{NH B}) &= \exp(\hat{\beta}_3 - \hat{\beta}_4)\end{aligned}$$

Reference: Using `estimable()` in Option 2

$$\hat{\beta}_3 - \hat{\beta}_4$$

- We could use `estimable()` to calculate this linear combination of coefficients (see [BSTA 512 Lesson](#))

```

1 library(gmodels)
2 RE_glm %>% estimable(
3   c("(Intercept)"
4     "Race_EthnicityHispanic-Latino"
5     "Race_EthnicityNH American Indian/Alaskan Native"
6     → "Race_EthnicityNH Asian/Pacific Islander"
7     → "Race_EthnicityNH Black"
8     conf.int = 0.95) %>%
9   exp(.)

```

	Estimate	Std. Error	X^2 value	DF	Pr(> X^2)	Lower.CI	Upper.CI
(0 0 0 1 -1)	0.8051811	1.107021	93.89366	2.718282	1.033623	0.6580966	0.9851389

$\begin{aligned} &= 0, & \# \text{ beta0} \\ &= 0, & \# \text{ beta1} \\ &= 0, & \# \text{ beta2} \\ &= 1, & \# \text{ beta3} \\ &= -1, & \# \text{ beta4} \end{aligned}$

1 → $\beta_3 - \beta_4$

The estimated odds of LSBC
 diagnosis for NH Asian/Pacific
 Islander individuals is 0.805
 times that of NH Black individuals
 (95% CI: 0.658, 0.985)

$$\begin{aligned}
 &0 \cdot \hat{\beta}_0 + 0 \hat{\beta}_1 + \\
 &0 \hat{\beta}_2 + 1 \hat{\beta}_3 + \\
 &(-1) \hat{\beta}_4
 \end{aligned}$$

Poll Everywhere Question 5

What if you want to compare other groups? Option 1

```
1 bc3 = bc %>%
2   mutate(Race_Ethnicity = relevel(Race_Ethnicity, ref = "NH Black"))
3 RE_glm2 = glm(Late_stage_diag ~ Race_Ethnicity, data = bc3,
4               family = binomial)
5 tidy(RE_glm2, conf.int=T, exponentiate = T) %>% gt() %>%
6   tab_options(table.font.size = 38) %>%
7   fmt_number(decimals = 3)
```

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	0.523	0.065	-9.934	0.000	0.459	0.594
Race_EthnicityNH White	0.712	0.070	-4.826	0.000	0.621	0.818
Race_EthnicityHispanic-Latino	0.689	0.102	-3.664	0.000	0.564	0.840
Race_EthnicityNH American Indian/Alaskan Native	0.675	0.479	-0.819	0.413	0.242	1.641
Race_EthnicityNH Asian/Pacific Islander	0.805	0.102	-2.131	0.033	0.659	0.982

Learning Objectives

1. Interpret odds ratios from fitted simple logistic regression model for a continuous explanatory variable.
2. Interpret odds ratios from fitted simple logistic regression model for a binary explanatory variable.
3. Interpret odds ratios from fitted simple logistic regression model for a multi-level categorical explanatory variable.
4. Report the odds ratio using a table and/or a forest plot.

How to present odds ratios: Table

- `tbl_regression()` in the `gtsummary` package is helpful for presenting the odds ratios in a clean way

```
1 library(gtsummary)
2 tbl_regression(RE_glm, exponentiate = TRUE) %>%
3   as_gt() %>% # allows us to use tab_options()
4   tab_options(table.font.size = 38)
```

Characteristic	OR ¹	95% CI ¹	p-value
Race_Ethnicity			
NH White	—	—	
Hispanic-Latino	0.97	0.82, 1.14	0.7
NH American Indian/Alaskan Native	0.95	0.34, 2.29	>0.9
NH Asian/Pacific Islander	1.13	0.96, 1.33	0.13
NH Black	1.40	1.22, 1.61	<0.001

¹ OR = Odds Ratio, CI = Confidence Interval

How to present odds ratios: Forest Plot Setup

```
1 library(broom.helpers)
2 RE_tidy = tidy_and_attach(RE_glm, conf.int=T, exponentiate = T) %>%
3   tidy_remove_intercept() %>%
4   tidy_add_reference_rows() %>% tidy_add_estimate_to_reference_rows() %>%
5   tidy_add_term_labels()
6 glimpse(RE_tidy)
```

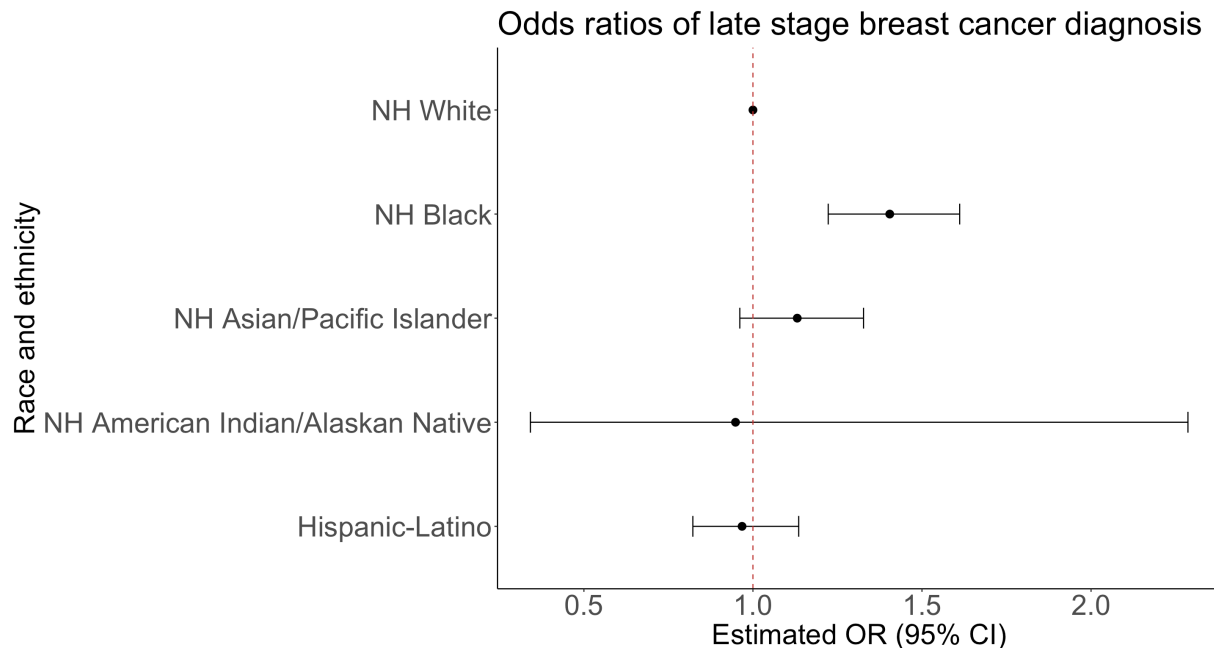
Rows: 5

Columns: 16

\$ term	<chr> "Race_EthnicityNH White", "Race_EthnicityHispanic-Latin...
\$ variable	<chr> "Race_Ethnicity", "Race_Ethnicity", "Race_Ethnicity", "...
\$ var_label	<chr> "Race_Ethnicity", "Race_Ethnicity", "Race_Ethnicity", "...
\$ var_class	<chr> "factor", "factor", "factor", "factor", "factor"
\$ var_type	<chr> "categorical", "categorical", "categorical", "categoric...
\$ var_nlevels	<int> 5, 5, 5, 5, 5
\$ contrasts	<chr> "contr.treatment", "contr.treatment", "contr.treatment"...
\$ contrasts_type	<chr> "treatment", "treatment", "treatment", "treatment", "tr...
\$ reference_row	<lgl> TRUE, FALSE, FALSE, FALSE, FALSE
\$ label	<chr> "NH White", "Hispanic-Latino", "NH American Indian/Alas...
\$ estimate	<dbl> 1.0000000, 0.9678002, 0.9484848, 1.1310170, 1.4046741
\$ std.error	<dbl> NA, 0.08224948, 0.47558680, 0.08224988, 0.07041472
\$ statistic	<dbl> NA, -0.3979312, -0.1112089, 1.4968682, 4.8257715
\$ p.value	<dbl> NA, 6.906809e-01, 9.114507e-01, 1.344276e-01, 1.394623e...
\$ conf.low	<dbl> NA, 0.8223138, 0.3417844, 0.9612074, 1.2226824
\$ conf.high	<dbl> NA, 1.135332, 2.286596, 1.327092, 1.611466

How to present odds ratios: Forest Plot

```
1 ggplot(data=RE_tidy, aes(y=label, x=estimate, xmin=conf.low, xmax=conf.high)) +  
2   geom_point(size = 3) +   geom_errorbarh(height=.2) +  
3   geom_vline(xintercept=1, color='#C2352F', linetype='dashed', alpha=1) +  
4   theme_classic() +  
5   labs(x = "Estimated OR (95% CI)", y = "Race and ethnicity",  
6         title = "Odds ratios of late stage breast cancer diagnosis") +  
7   theme(axis.title = element_text(size = 25), axis.text = element_text(size = 25), title =
```



References

- Mateo, Camila M., and David R. Williams. 2021. "Racism: A Fundamental Driver of Racial Disparities in Health-Care Quality." *Nature Reviews Disease Primers* 7 (1): 1–2. <https://doi.org/10.1038/s41572-021-00258-1>.
- Yedjou, Clement G., Jennifer N. Sims, Lucio Miele, Felicite Noubissi, Leroy Lowe, Duber D. Fonseca, Richard A. Alo, Marinelle Payton, and Paul B. Tchounwou. 2019. "Health and Racial Disparity in Breast Cancer." *Advances in Experimental Medicine and Biology* 1152: 31–49. https://doi.org/10.1007/978-3-030-20301-6_3.